Advanced Empirical Finance: Topics and Data Science

Asset pricing and reproducibility

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Asset pricing models

The fundamental equation

$$0 = E(m\,r)$$

- A linear SDF $m = a - b'\,f$ implies **beta pricing**: $E(r_i) = \beta_i'\,\lambda$, where $\beta_i = \mathrm{Cov}(r_i, f)/\mathrm{Var}(f)$ are factor loadings and λ are factor risk premia

Multifactor asset pricing models

- FF (1992, 1993) have shown that the following factors capture parts of the cross-sectional variation in expected returns:

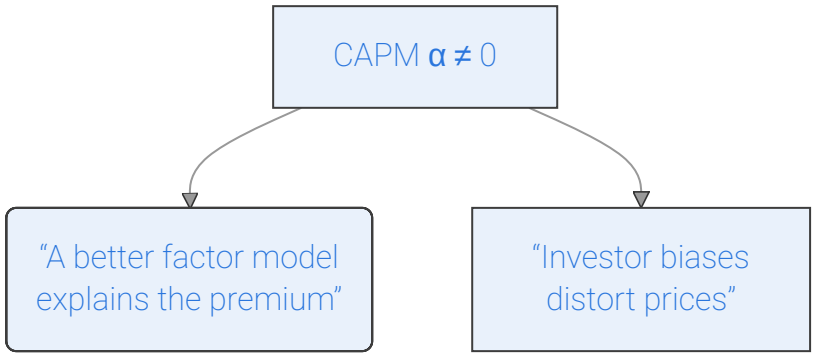
1. the return on the market portfolio in excess of the risk-free rate of return
2. a zero net investment (spread) portfolio long in small firm stocks and short in large firm stocks (SMB)
3. a spread portfolio long in high book-to-market stocks and short in low book-to-market stocks (HML)

- Basic CAPM risk-return trade-off: high risk comes along with a high expected return
- Conditional versions: risk and risk premium could vary with macro economy variables such as the default spread
- Ross (1976) introduced the Arbitrage Pricing Theory (APT) as an alternative to the Capital Asset Pricing Model

Market anomalies

Anomaly	Finding	Key reference(s)
Betting against beta	α declining in β across many asset classes	Frazzini & Pedersen (2014)
Size	Small stocks outperform large stocks	Banz (1981); FF (1992)
Value	High book-to-price stocks outperform growth stocks	Rosenberg, Reid & Lanstein (1985); FF (1992)
Momentum	Intermediate-term winners outperform losers	Jegadeesh & Titman (1993); Carhart (1997)
Time-series momentum	Assets trending up (down) continue to do so	Moskowitz, Ooi & Pedersen (2012)
Liquidity	Illiquid stocks earn a return premium	Pastor & Stambaugh (2003)

Given these anomalies: Are markets efficient?



- Cochrane (2011): we can rationalize any arbitrage-free data with $0 = E_t(m_{t+1}r_{t+1})$
- but a model only gets its bite by restricting m or distorted expectations, ideally tied to other data

Arbitrage Pricing Theory

- APT assumes that markets are competitive and frictionless and that the return-generating process is

$$R_i = a_i + b_i' f + \varepsilon_i$$

where R_i is the return for asset i, a_i is the intercept of the factor model, b_i is a $(K \times 1)$ vector of factor sensitivities (loadings) and f is a $(K \times 1)$ vector of common factor realizations

- **Note:** f is not necessarily a tradeable risk factor
- For the system of N assets we have

$$R = a + Bf + \varepsilon$$

where B is a $(N \times K)$ matrix of factor loadings and $\mathrm{Cov}(\varepsilon) = \Sigma$

- Estimation is straightforward if the factors f are tradeable portfolios

APT (Ross, 1976)

- The absence of arbitrage implies

$$E(R) = \mu = \iota\lambda_0 + B\lambda_k$$

where λ_0 is the zero-beta parameter (usually the risk-free rate) and λ_k is the $(K \times 1)$ vector of risk premia. Then, the estimation rests on the regression

$$Z_t = a + BZ_{K,t} + \varepsilon_t$$

where Z_t are excess returns

- All the exact factor pricing models allow one to estimate the expected return on a given asset
- One needs measures of the factor sensitivity matrix B, the risk-free rate or the zero-beta expected return, and the factor risk premia λ_k
- In the case where the factors are the excess returns on traded portfolios, we can estimate the risk premia directly from the sample means of the excess returns on the portfolios

Open question: How to choose the factors?

Considerations

- Models may overfit the data because of data-snooping bias; in this case, they will not be able to predict asset returns in the future (you can always add another factor to increase in-sample goodness of fit)
- Models may capture empirical regularities that are due to market inefficiency or investor irrationality; in this case, they may continue to fit the data but they will imply Sharpe ratios for factor portfolios that are too high to be consistent with a reasonable underlying model of market equilibrium

Factor selection: Principal Components

- Well-known statistical procedure: Principal components
- Principal components is a technique to reduce the number of variables we study without losing too much information in the covariance matrix
- The principal components serve as the factors. The first principal component is the (normalized) linear combination of asset returns with maximum variance
- Let the $(N \times 1)$ vector x_1^* be the solution to

$$\max_{x_1} x_1' \hat{\Omega} x_1 \quad s.t. \quad x_1' x_1 = 1$$

where $\hat{\Omega}$ is the sample variance-covariance matrix of the returns

- x_1^* is the eigenvector associated with the largest eigenvalue of $\hat{\Omega}$
- For principal component x_2^*, solve the same optimization problem with the additional constraint that $x_2' x_1^* = 0$
- Partial least squares perform dimension reduction by directly exploiting covariation of predictors with the forecast target

Factor selection: Theoretical approach

- Theoretical approach: identify variables that capture systematic economy-wide risks
  - → *Macro variables*: interest rates, default spreads, inflation — proxies for systematic risk sources
  - → *Firm characteristics*: size, book-to-market, momentum — proxies for differential risk exposure; sort stocks into portfolios accordingly
- For more information, consider Chapter 6 of "The Econometrics of Financial Markets" (Campbell, Lo & MacKinlay, 1997)

Bayesian mindset: don't compromise on a single model

- Assess the probability that a candidate model generates the observed dynamics of asset returns
- Integrate over factor models, weighting them by their posterior probabilities
- Approach follows directly from the Bayes rule: integrated factor model summarizes the uncertainties about the joint dynamics of stock returns

"Integrating Factor Models", Avramov et al. (2023) and "Bayesian Solutions to the Factor Zoo", Bryzgalova et al. (2023)

- Flexible to consider models with time-varying parameters
- Prior beliefs about the entire parameter space are economically interpretable and weigh against model mispricing and the inclusion of macro predictors
- Posterior probability weighs against deviations from the unconditional CAPM
- Penalize model complexity, i.e., an incremental factor, beyond the market, or macro predictor stays in the model only if it considerably improves model pricing abilities

Integrating Factor Models

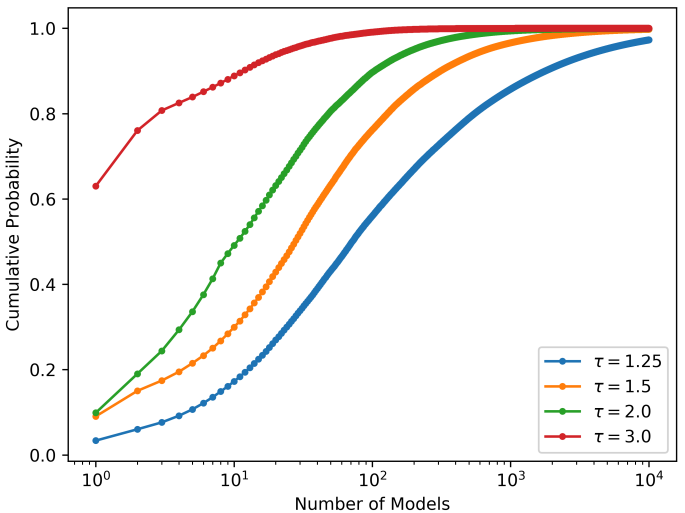
- Excess returns: multivariate asset pricing regression

$$r_{t+1} = \alpha(z_t) + \beta(z_t) f_{t+1} + u_{r,t+1}, u_{r,t+1} \sim N(0, \Sigma_{RR})$$

- Factors: multivariate predictive regression

$$f_{t+1} = \alpha_f + a_F z_t + u_{f,t+1}, u_{f,t+1} \sim N(0, \Sigma_{FF})$$

- r_{t+1}: N-vector of excess returns
- f_{t+1}: K-vector of factors
- z_t: M-vector of macro predictors
- $\alpha(z_t) = \alpha_0 + \alpha_1 z_t$, fixed and time-varying mispricing
- $\beta(z_t) = \beta_0 + \beta_1 (I_K \otimes z_t)$, fixed and time-varying factor loadings
- a_F captures time-varying risk premia



From model proliferation to reproducibility

- The factor zoo raises a natural question: which findings are genuine?
- The remainder of this lecture examines why many published results fail to replicate — and what we can do about it

The reproducibility crisis

Reproducibility

- Research should be reproducible — enforcement becomes increasingly critical
- What does reproducible mean? Would you say your reports are reproducible?

Collect ideas: Is empirical research in finance reproducible?

- What are limiting factors?
- What problems arise with non-reproducible research (in academia and finance)?

Minimum requirements for reproducibility

1. A set of files with the raw data and code. It is possible to create the tables and any data-derived charts/graphics/visualizations by running the code
2. Details about the system that runs the analysis: you list the operating system, patches, random number seeds (`set.seed(3010)`), and specific versions of all software/packages/libraries
3. Open/transparent. All the data and materials are available (as opposed to "available upon request") – e.g., you post them on GitHub
4. You write code that others can readily understand

Code & data sharing policies in top journals



- **Code**: journals require it for empirical work, simulations, numerical computations and experimental studies
- **Data**: if authors cannot share actual data (due to legal/confidentiality issues), they must provide pseudo-data

Reproducibility also important in industry

- **Transparency**: clear code & data practices build client & regulatory trust
- **Collaboration**: well-documented workflows enable smooth team handovers
- **Maintainability**: clean processes simplify updates & scaling
- **Faster Innovation**: reproducible processes accelerate testing & deployment

How empirical projects sometimes feel



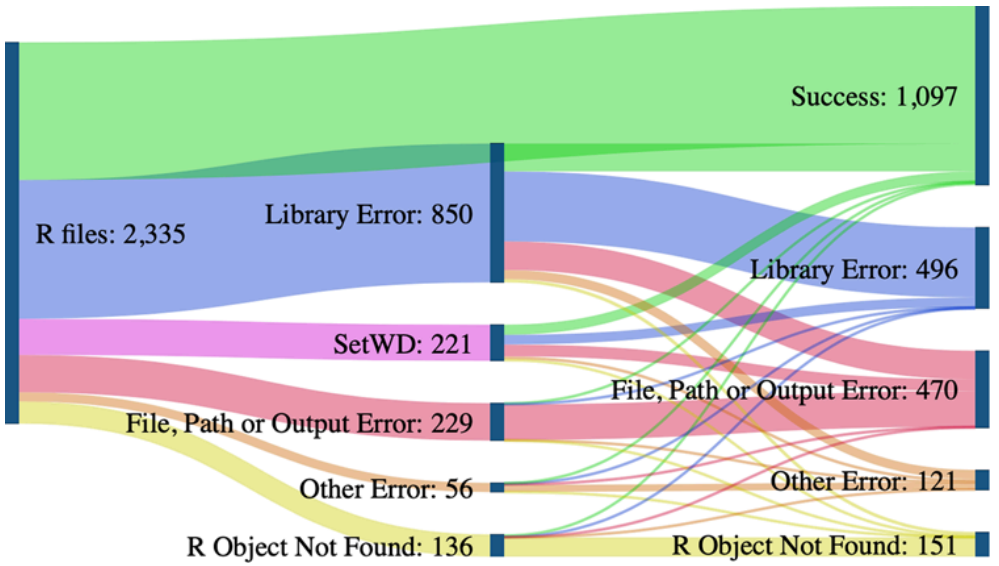


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What can go wrong?

Key takeaways

- Create a **separate project** for each data analysis
- Save & organize scripts with **descriptive names**
- Regularly **restart your session** to ensure your scripts capture all work
- Always use **relative paths**, never absolute paths
- Avoid **manual steps** as much as possible



Research code quality assessment of Harvard Dataverse between 2010-2020. Source: @trisovic2022

The replication crisis of finance

"... most claimed research findings in financial economics are likely false."



Campbell Harvey
in Harvey, Liu, and Zhu (2016, RFS)

"We develop and estimate a Bayesian model of factor replication that leads to different conclusions."



Bryan Kelly
in Jensen, Kelly, Pedersen (2023, JF)

P-hacking: What is odd with this figure?

- Publication bias: reported t-stats of asset pricing *factors*

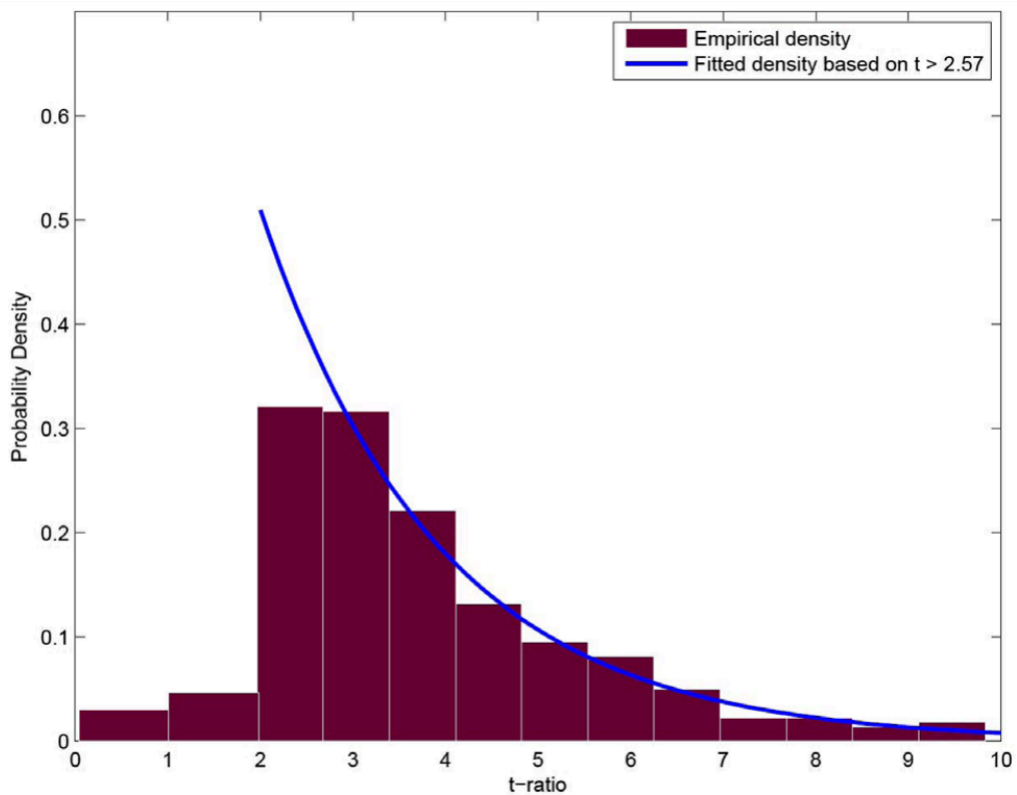
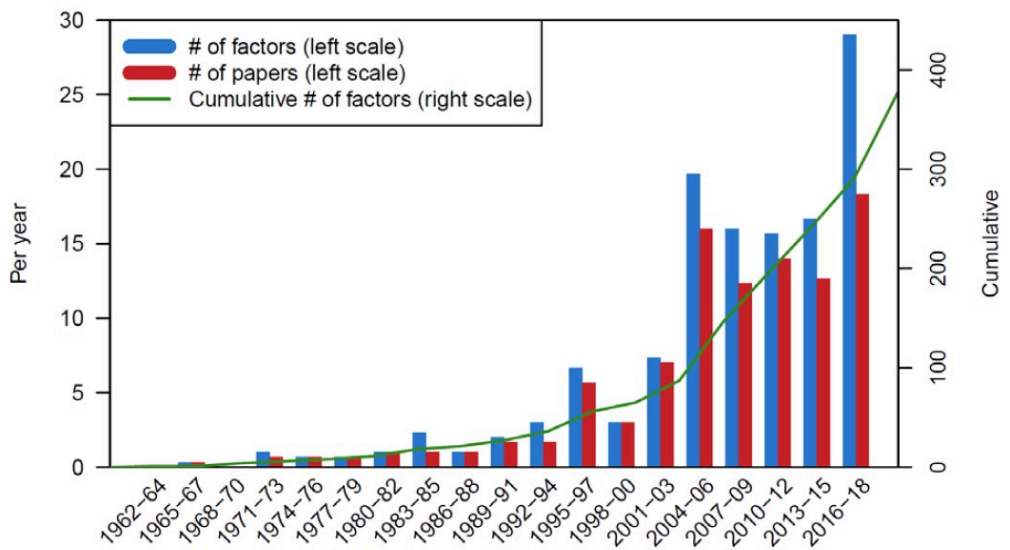


Figure 1. Distribution of reported *t*-statistics from factor studies, 1963 to 2012. Evidence from Harvey, Liu, and Zhu (2016).

- Studies cluster just above the conventional significance thresholds (Harvey, 2017): the spike just above $t = 1.96$ and $t = 2.57$ is far larger than a smooth distribution would predict
- Notice very few papers reporting results below conventional significance levels

The factor zoo

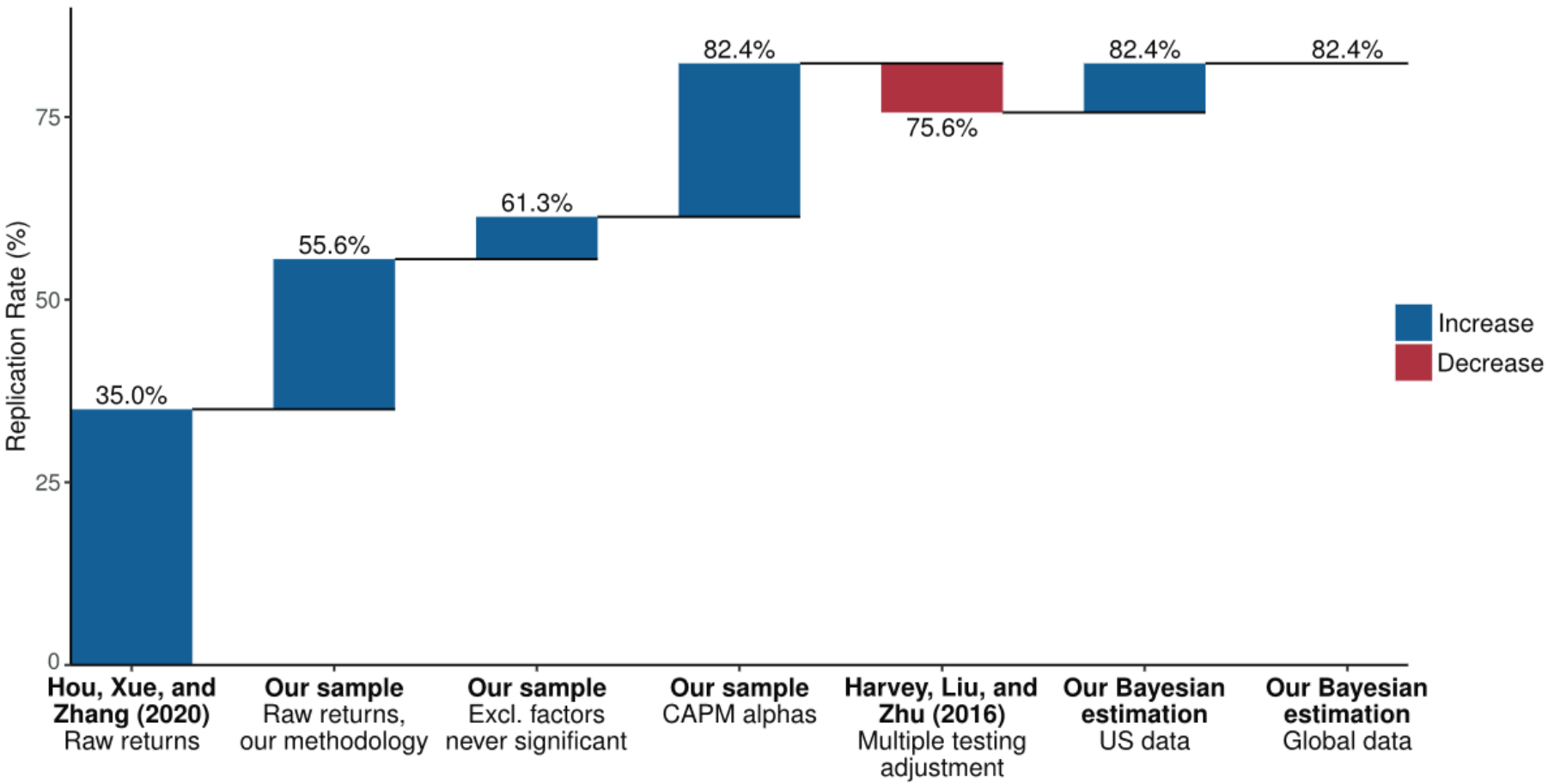
Figure 1. Out of control factor production - through January 2019



* Journals published through December 2018. Data collection in January 2019.

- Cochrane (2011): "Now we have a zoo of new factors"
- Do we really believe in 300+ risk factors?
- Machine learning's aim: tame the factor zoo
- See **this site** for an updated sheet with published factors and some descriptions

Jensen et al (2023): Is there a replication crisis in Finance?



- Replication is not just about running the same code — it depends critically on what question you ask and how you measure the answer
- Bayesian reasoning lets you pool evidence across many related tests, which is both more principled and more powerful than applying ad hoc frequentist corrections

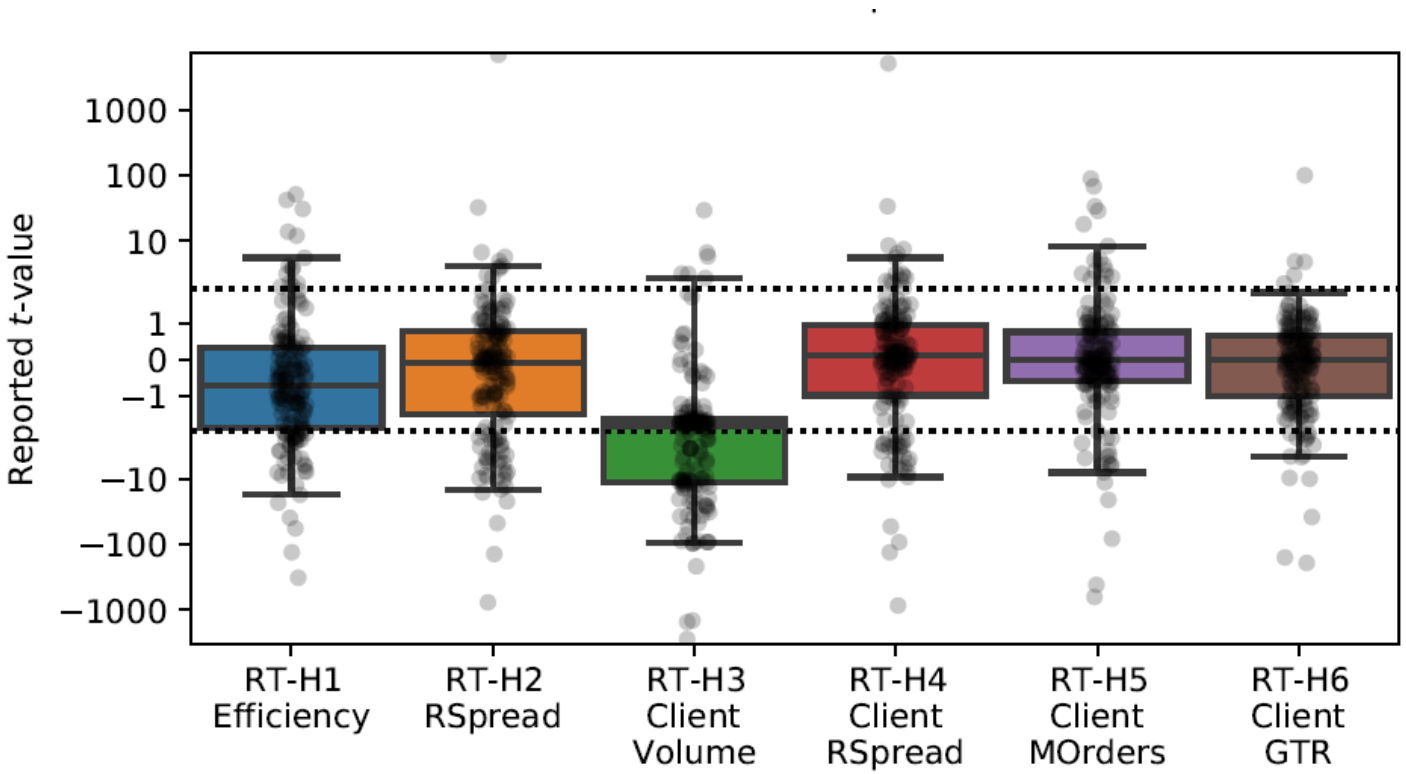
Related: Non-standard errors

- Albert J. Menkveld led a large research project (**Paper available here**)
- Uncertainty from the **evidence generating process** (EGP)
- EGP variation across researchers adds uncertainty: non-standard errors
- 164 teams test six hypotheses on the same sample

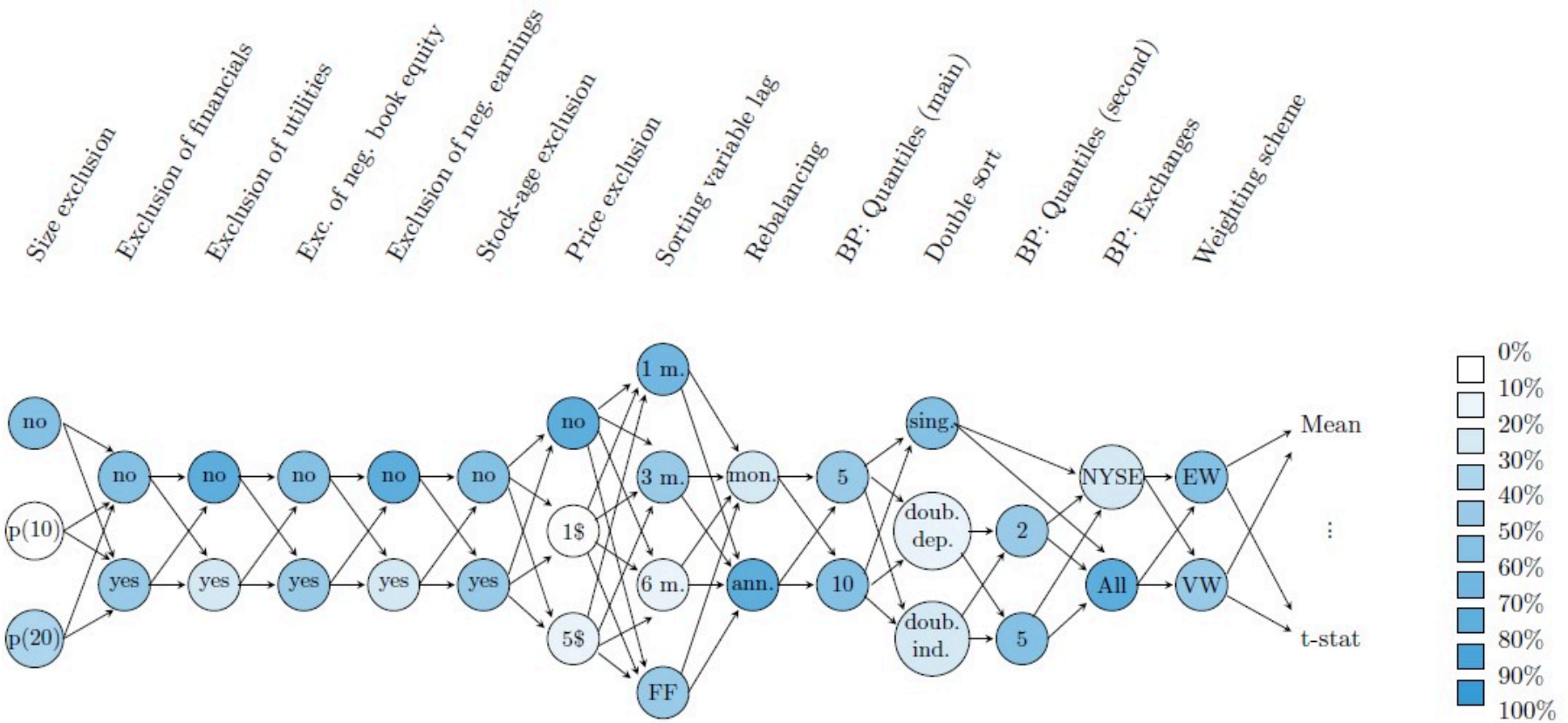


Related: Non-standard errors

- Non-standard errors are sizeable, on par with standard errors
- Their size (i) co-varies only weakly with team merits, reproducibility, or peer rating, (ii) declines significantly after peer feedback, and (iii) far exceeds what participants expect

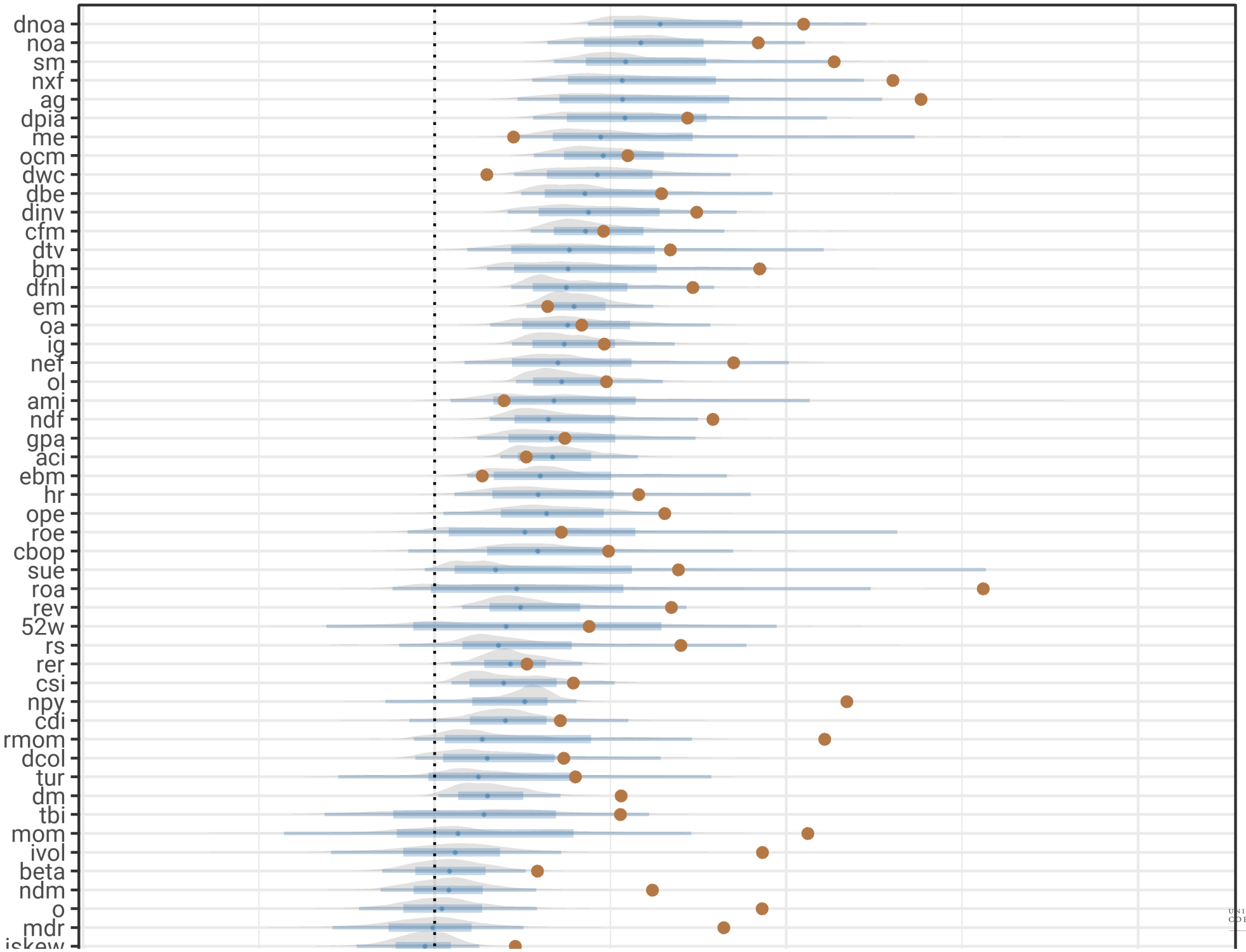


Non-standard errors in portfolio sorts



- Sample construction: exclusions based on market equity, industry, firm-months with negative book equity or negative earnings, stock-age or stock price
- Portfolio construction: lag of the sorting variables, portfolio rebalancing frequency, the number of portfolios, listing exchanges for breakpoints, weighting scheme
- Color saturation indicates how often the 109 papers that Hou et al. (2020) analyzed implemented each choice.

Uncertainty everywhere



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Data preprocessing choices matter

A simple example

- Suppose the SDF is linear in the market and 50 other risky factors (we fix f)
- Now: The *idea* of a risk factor remains the same, but the *data preprocessing choices* ($\mathcal{D}$) vary
- We consider 3,840 plausible alternatives for portfolio sorts for *each* sorting variable
- Plethora of different ways to represent an otherwise identical SDF
- Data preprocessing choices alter the perceived efficient frontier and thus affect investment decisions
- Ignoring conceptual uncertainty implies a very strong prior

